

CENTRAL AGGREGATION SERVER (Cloud / Orchestrator)

State Update λ_{t+1} & Reward Signal R_t

Level-2 Client Selector

- Multi-armed Bandit Policy $\pi(S_t|X_t)$
- State X_t : [Loss, Acc, Latency, Energy]

Poisoning & Anomaly Filter

- Cosine Gradient Similarity
- Statistical Z-Score Thresholding
- Malicious Update Rejection

Level-1 Meta-Aggregator

- Strategy Selection (FreeAvg / V-Prox)
- Global Model Update: $W_{t+1} \leftarrow W_t + \Delta W$
- Global Performance Evaluation

Downlink:
Selection S_t &
Global Weights W_t

Uplink:
Updates w_i^{t+1} &
Metrics (L_i, T_i, E_i)

HETEROGENEOUS EDGE CLIENT CLUSTER (N Clients)

Tier 1: Workstation

Hardware: 2.5 GHz | 8.0W | High BW

- Local ECG Partition D_i
- CNN Trainer (E Epochs)
- Energy & Latency Profiling

Tier 2/3: Mobile/Tablet

Hardware: 1.8-1.2 GHz | 4.5-2.5W | Mid BW

- Local ECG Partition D_i
- CNN Trainer (E Epochs)
- Energy & Latency Profiling

Tier 4: IoT Node

Hardware: 0.8 GHz | 1.2W | Low BW

- Local ECG Partition D_i
- CNN Trainer (E Epochs)
- Energy & Latency Profiling